Week 2 Penetration Testing Report

Footprinting and Network Scanning

Program	Networkwalks Cybersecurity Internship
Week	02
Modules	W2-PM3 Maltego Footprinting, W2-PM5 Zenmap Network Scanning, W2-PM-FINAL Report
Author	Dhairay Kanojiia
Batch ID	B084
Internship ID	NW-084-001
Date of practicals	09 Nov 2026 (Maltego), 09 Nov 2026 (Zenmap)
Environment	Kali Linux in VirtualBox

1. Liability Disclaimer

These activities were performed only on public information, on systems I own, or inside my own VirtualBox lab network. All material is for educational purposes. Unauthorized access to systems or networks is illegal. Personal names and phone numbers that appeared in the Maltego results have been redacted in this report because they belong to private individuals.

2. Introduction and Objective

This report covers two practicals from Week 2 of the Networkwalks program: passive OSINT footprinting with Maltego (W2-PM3) and host discovery with Zenmap (W2-PM5). The objective was to understand how much information can be collected about a target before any exploitation, and to document the results with evidence.

3. Tools Used

Tool	Purpose
Kali Linux (VirtualBox)	Operating system used for all practicals
Maltego CE	Visual OSINT: turns a domain into a graph of related entities using transforms
ping	Check connectivity and latency to a host using ICMP
arp	Show the IP to MAC address mappings the system has learned
Zenmap (Nmap 7.991)	GUI for Nmap, used here for a ping scan (host discovery)

4. W2-PM3: Maltego Footprinting

Target: skrpublicschool.org. **Type:** passive OSINT, no packets were sent to the target's servers by me; Maltego transforms query public sources (WHOIS and search results).

4.1 Theory

Footprinting is the first phase of a security assessment, where information about a target is collected. Passive footprinting uses public sources only. Active footprinting touches the target directly (for example a port scan). Maltego works with **entities** (domain, person, email, phone, website, document) and **transforms**, which take one entity and return related ones. The result is a graph that shows relationships.

4.2 Steps performed

• Launched Maltego CE on Kali Linux (Figure 1).

• Created a new graph and added a Domain entity: skrpublicschool.org.

• Ran transforms on the domain and on the entities it returned.

• Reviewed the graph in entity view (Figure 2) and bubble view (Figure 3), then saved the graph.



Figure 1: Maltego CE starting on Kali Linux

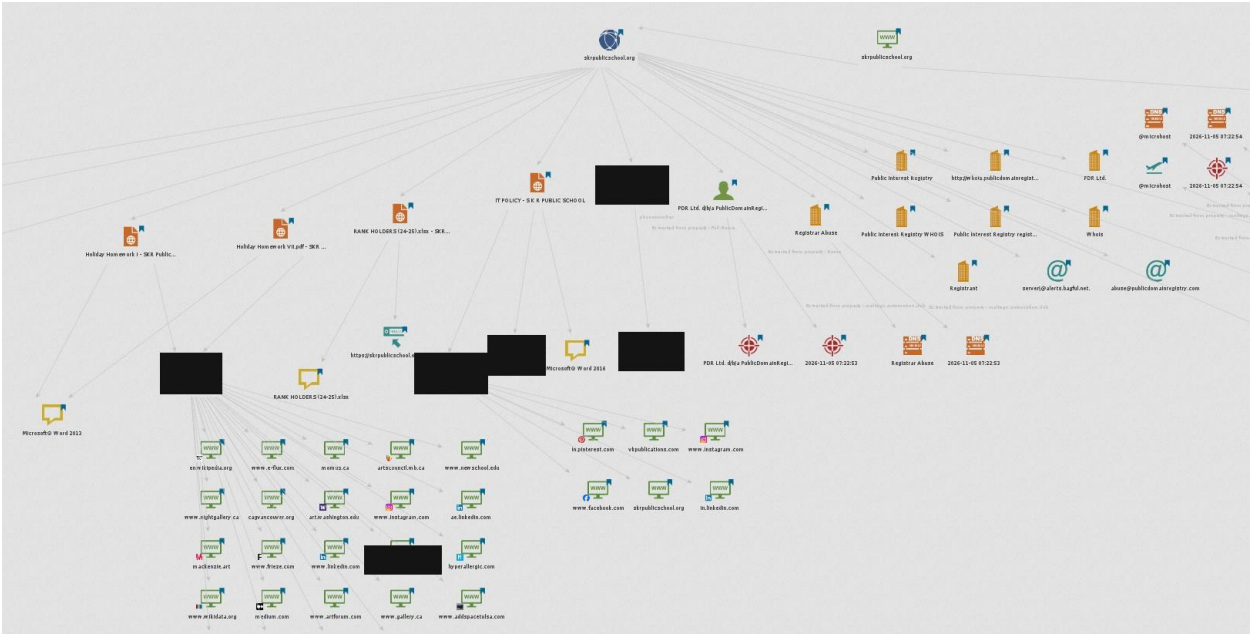


Figure 2: Entity view of the graph for skrpublicschool.org (personal names and phone numbers redacted)

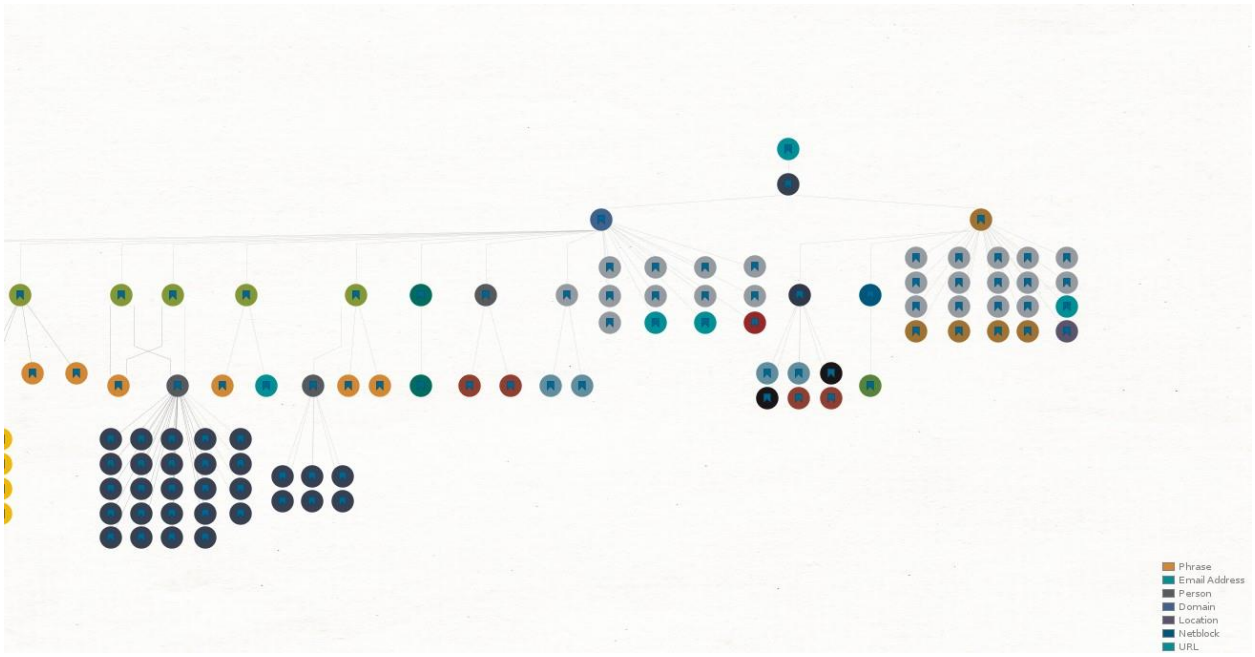


Figure 3: Bubble view of the same investigation. Legend: Phrase, Email Address, Person, Domain, Location, Netblock,

URL

4.3 Findings

- **Registration data:** the graph links the domain to Public Interest Registry (the .org registry) and to the registrar PDR Ltd. d/b/a PublicDomainRegistry.com, plus registrar abuse contact entities and the registrar abuse email address.
- **Emails:** only registrar and server-side addresses appeared (abuse contact and an alerts mailbox on a hosting-related domain). No staff personal email was shown in this graph.
- **Documents:** public files hosted on the domain (a holiday homework PDF, a rank holders spreadsheet and an IT policy document) were picked up. Their metadata produced Person entities (document authors), software names (Microsoft Word 2013 and 2016) and a phone number.
- **Websites and social links:** related websites and social profiles (Instagram, Facebook, LinkedIn, Pinterest and others) were linked to the person entities.
- **DNS entities** with timestamps from 2026-11-09 were created during the transform run.

4.4 Observation

Metadata inside public documents leaks author names and software versions, and a few transforms are enough to connect those names to social media. This is an observation from the graph, not a vulnerability test. Nothing was accessed or exploited.

5. W2-PM5: Zenmap Network Scanning

Environment: my own VirtualBox lab network 10.0.2.0/24. **Target:** 10.0.2.3 (a host in the lab network). No port scan was run, only host discovery.

5.1 Theory

Nmap sends probes to find which hosts are alive. A **ping scan** (`-sn`) only checks if a host is up and does not scan ports. Zenmap is the graphical front end of Nmap. On a local network, Nmap can also read the target's MAC address through ARP.

5.2 Step 1: ICMP connectivity check

Command: `ping -c 4 10.0.2.3`

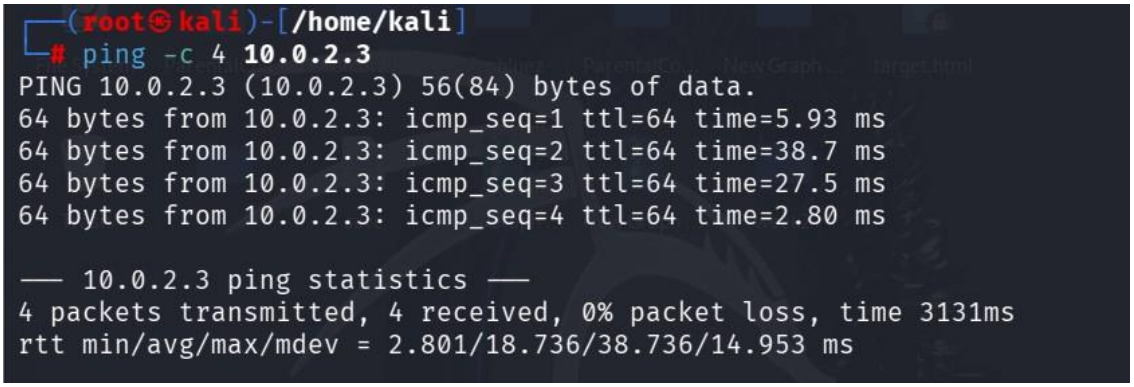


Figure 4: ping to 10.0.2.3

- 4 packets sent, 4 received, 0% packet loss.
- TTL 64, which is the default for Linux-based systems (an indication, not proof).
- Round trip time: min 2.801 ms, avg 18.736 ms, max 38.736 ms.

5.3 Step 2: ARP table

Command: `arp`



Figure 5: ARP table on the Kali machine

IP address	MAC address	Note
10.0.2.1	[redacted]	Network gateway/virtual router of the lab network
10.0.2.3	[redacted]	Target host. The 08:00:27 prefix is the Oracle VirtualBox vendor prefix

5.4 Step 3: Zenmap ping scan

Zenmap profile: **Ping scan**. Command: `nmap -sn 10.0.2.3`

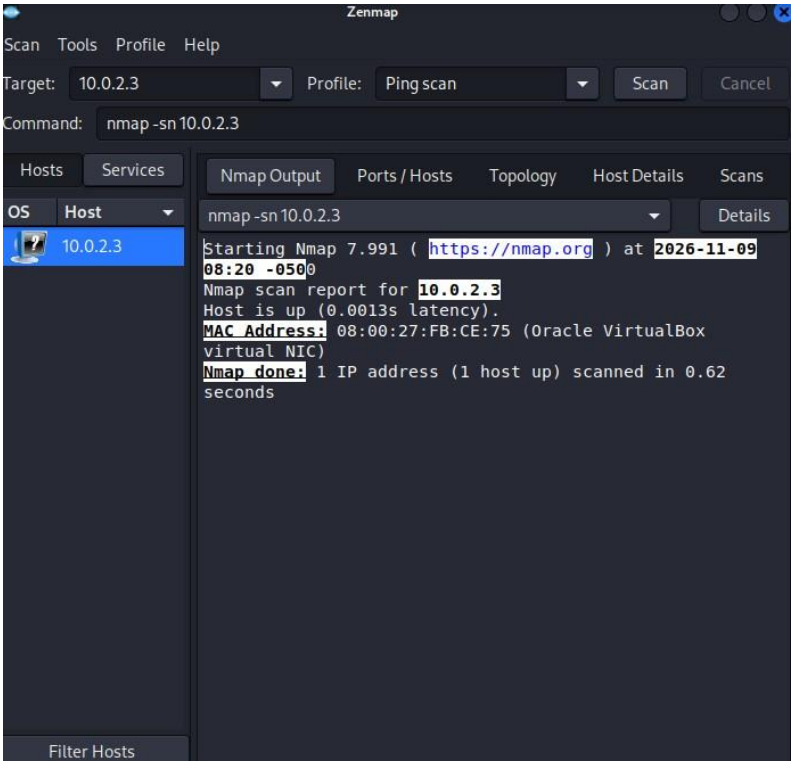


Figure 6: Zenmap ping scan result

• Scan started 2026-11-09 08:20 (Nmap 7.991).

• Host is up, latency 0.0013 s.

• MAC address [redacted] identified as an Oracle VirtualBox virtual NIC.

• 1 IP address scanned, 1 host up, finished in 0.62 seconds.

6. Findings and Observations

#	Observation	Evidence	Impact	Level
1	Metadata in public documents exposes author names and software versions	Maltego graph, person and phrase entities	Gives attackers names for social engineering	○ Medium
2	Registrar and abuse contacts are publicly visible through WHOIS	Maltego graph, registrar and email entities	Normal for domain registration, low risk on its own	○ Low
3	Personal profiles are linkable to the organization's documents	Maltego graph, website entities	Helps build a profile of individuals	○ Medium
4	Target host replies to ping and ARP	Figures 4, 5, 6	Host is easy to discover on the local network	○ Low
5	MAC vendor prefix reveals virtualization	Figures 5, 6	Shows the host is a VirtualBox VM	○ Low

These are observations from information gathering, not confirmed vulnerabilities. No exploitation or vulnerability validation was done.

7. Recommendations

- Remove metadata (author, software, properties) from documents before publishing them on awebsite.
- Review which files are public on the domain and remove those that do not need to be public.
- Use WHOIS privacy where the registrar allows it.
- Keep personal social media separate from organization documents.
- Run host discovery on your own network regularly and investigate unknown devices.
- Only perform scans and OSINT on targets where you have authorization.

8. Conclusion

In W2-PM3 I used Maltego to map the public footprint of a domain and saw how document metadata, WHOIS data and web results connect into one picture. In W2-PM5 I used ping, ARP and Zenmap on my own lab network to discover a host, read its MAC address and confirm it was a VirtualBox VM. Together they show the first two phases of an assessment: gathering information without touching the target, and then discovering live hosts. Both were done inside the intended scope and documented with screenshots.